

DQL Queries for Kubernetes

Series: K8S | **Notebook:** 8 of 9 | **Created:** January 2026

Advanced Query Patterns for Kubernetes Data

This notebook provides a comprehensive reference of DQL queries for Kubernetes monitoring. From basic entity queries to complex performance analysis, these patterns help you extract insights from your Kubernetes data.

Table of Contents

1. Entity Queries
 2. Metric Queries
 3. Log and Event Queries
 4. Trace Queries
 5. Correlation Queries
 6. Dashboard Queries
 7. Alerting Queries
 8. Query Templates
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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Kubernetes monitoring
Permissions	<code>metrics.read</code> , <code>entities.read</code> , <code>logs.read</code>
Knowledge	K8S-01 through K8S-07
Data	Active Kubernetes cluster monitored

1. Entity Queries

Kubernetes Entity Types

Entity Type	DQL Table	Description
Cluster	<code>dt.entity.kubernetes_cluster</code>	K8s clusters
Node	<code>dt.entity.kubernetes_node</code>	Worker nodes
Namespace	<code>dt.entity.cloud_application_namespace</code>	Namespaces
Workload	<code>dt.entity.cloud_application</code>	Deployments, StatefulSets
Container	<code>dt.entity.container_group_instance</code>	Container instances
Service	<code>dt.entity.service</code>	Detected services

```
// List all Kubernetes clusters
fetch dt.entity.kubernetes_cluster
| fields entity.name, clusterId, cloudType, tags
| sort entity.name asc
```

```
// List all nodes with cluster relationship
fetch dt.entity.kubernetes_node
| fields entity.name, kubernetesClusterName, cpuCores, physicalMemory
| sort kubernetesClusterName asc, entity.name asc
```

```
// List all namespaces
fetch dt.entity.cloud_application_namespace
| fields entity.name, kubernetesClusterName
| sort kubernetesClusterName asc, entity.name asc
```

```
// List all workloads (deployments, statefulsets)
fetch dt.entity.cloud_application
| fields entity.name, cloudApplicationNamespaces, workloadKind
| sort cloudApplicationNamespaces asc, entity.name asc
| limit 50
```

```
// Count entities by type per cluster
fetch dt.entity.kubernetes_node
| summarize nodes = count(), by:{kubernetesClusterName}
| sort nodes desc
```

2. Metric Queries

Common Kubernetes Metrics

Metric	Description	Unit
<code>dt.containers.cpu.usage_percent</code>	Container CPU vs limit	Percent
<code>dt.containers.memory.usage_percent</code>	Container memory vs limit	Percent
<code>dt.containers.cpu.throttled_time</code>	CPU throttle time	Nanoseconds
<code>dt.containers.memory.working_set_bytes</code>	Working memory	Bytes
<code>dt.kubernetes.workload.requests_cpu</code>	CPU requests	Millicores
<code>dt.kubernetes.workload.requests_memory</code>	Memory requests	Bytes

```
// Container CPU usage – top consumers
timeseries cpu = avg(dt.containers.cpu.usage_percent), by:
{dt.entity.container_group_instance}
| sort avg(cpu) desc
| limit 15
```

```
// Container memory usage approaching limits
timeseries mem = avg(dt.containers.memory.usage_percent), by:
{dt.entity.container_group_instance}
| filter avg(mem) > 80
| sort avg(mem) desc
```

```
// CPU throttling detection
timeseries throttle = sum(dt.containers.cpu.throttled_time), by:
{dt.entity.container_group_instance}
| filter sum(throttle) > 0
| sort sum(throttle) desc
| limit 15
```

```
// Namespace-level resource usage
timeseries
  cpuUsage = avg(dt.containers.cpu.usage_percent),
  memUsage = avg(dt.containers.memory.usage_percent),
  by:{k8s.namespace.name}
| sort avg(cpuUsage) desc
| limit 15
```

```
// Resource requests by namespace (capacity planning)
timeseries cpuRequests = sum(dt.kubernetes.workload.requests_cpu), by:
{k8s.namespace.name}
| sort avg(cpuRequests) desc
| limit 15
```

3. Log and Event Queries

Log Query Patterns

```
// Error logs by namespace
fetch logs
| filter loglevel == "ERROR"
| filter isNotNull(k8s.namespace.name)
| summarize errorCount = count(), by:{k8s.namespace.name}
| sort errorCount desc
| limit 15
```

```
// Kubernetes events – warnings only
fetch logs
| filter matchesPhrase(content, "Warning")
| filter matchesPhrase(log.source, "kubernetes") or matchesPhrase(log.source, "k8s")
| fields timestamp, content
| sort timestamp desc
| limit 30
```

```
// Pod crashes and restarts
fetch logs
| filter matchesPhrase(content, "CrashLoopBackOff") or matchesPhrase(content, "BackOff")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// OOM events
fetch logs
| filter matchesPhrase(content, "OOMKilled") or matchesPhrase(content, "Out of memory")
| fields timestamp, content
| sort timestamp desc
| limit 20
```

```
// Log volume trend by namespace
fetch logs, from: now() - 24h
| filter isNotNull(k8s.namespace.name)
| summarize count(), by:{k8s.namespace.name, bin(timestamp, 1h)}
| sort timestamp asc
| limit 200
```

4. Trace Queries

Span Queries for Kubernetes Services

```
// Service response times in K8s
fetch spans
| filter span.kind == "server"
| filter isNotNull(k8s.namespace.name)
| summarize
    p50 = percentile(duration, 50),
    p95 = percentile(duration, 95),
    p99 = percentile(duration, 99),
    by:{k8s.namespace.name, dt.entity.service}
| sort p99 desc
| limit 15
```

```
// Error rate by service
fetch spans
| filter span.kind == "server"
| filter isNotNull(k8s.namespace.name)
| summarize
    total = count(),
    errors = countIf(otel.status_code == "ERROR"),
    by:{k8s.namespace.name, dt.entity.service}
| fieldsAdd errorRate = 100.0 * toDouble(errors) / toDouble(total)
| filter errors > 0
| sort errorRate desc
| limit 15
```

```
// Slow traces (>1 second)
fetch spans
| filter span.kind == "server" and duration > 1000000000
| filter isNotNull(k8s.namespace.name)
| fields timestamp, trace.id, k8s.namespace.name, span.name, duration
| sort duration desc
| limit 20
```

```
// Request throughput by namespace
timeseries requestRate = count(), by:{k8s.namespace.name}
| sort avg(requestRate) desc
| limit 10
```

5. Correlation Queries

Joining Multiple Data Sources

```
// High CPU workloads with their namespaces
timeseries cpu = avg(dt.containers.cpu.usage_percent), by:
{dt.entity.cloud_application}
| filter avg(cpu) > 50
| lookup [fetch dt.entity.cloud_application | fields id, entity.name,
cloudApplicationNamespaces], sourceField:dt.entity.cloud_application,
lookupField:id
| fields entity.name, cloudApplicationNamespaces, avg(cpu)
| sort avg(cpu) desc
```

```
// Nodes with high utilization
timeseries nodeCpu = avg(dt.kubernetes.node.cpu_usage), by:
{dt.entity.kubernetes_node}
| filter avg(nodeCpu) > 70
| lookup [fetch dt.entity.kubernetes_node | fields id, entity.name,
kubernetesClusterName], sourceField:dt.entity.kubernetes_node, lookupField:id
| fields entity.name, kubernetesClusterName, avg(nodeCpu)
| sort avg(nodeCpu) desc
```

6. Dashboard Queries

Queries Optimized for Dashboards

```
// Cluster summary tile
fetch dt.entity.kubernetes_cluster
| summarize clusterCount = count()
```

```
// Node count tile
fetch dt.entity.kubernetes_node
| summarize nodeCount = count()
```

```
// Error rate trend (chart)
fetch logs, from: now() - 24h
| filter loglevel == "ERROR" and isNotNull(k8s.namespace.name)
| summarize errors = count(), by:{bin(timestamp, 1h)}
| sort timestamp asc
```

```
// Top namespaces by CPU (bar chart)
timeseries cpu = avg(dt.containers.cpu.usage_percent), by:
{k8s.namespace.name}
| sort avg(cpu) desc
| limit 10
```

7. Alerting Queries

Threshold-Based Alerting Patterns

```
// High memory usage alert query
timeseries mem = avg(dt.containers.memory.usage_percent), by:
{dt.entity.container_group_instance}
| filter avg(mem) > 90
| summarize alertCount = count()
```

```
// CrashLoop detection (last hour)
fetch logs, from: now() - 1h
| filter matchesPhrase(content, "CrashLoopBackOff")
| summarize crashCount = count()
```

```
// Error rate spike detection
fetch logs, from: now() - 1h
| filter loglevel == "ERROR" and isNotNull(k8s.namespace.name)
| summarize errors = count(), by:{k8s.namespace.name}
| filter errors > 100
| sort errors desc
```


8. Query Templates

Reusable Query Patterns

Filter by Namespace:

```
| filter k8s.namespace.name == "NAMESPACE_NAME"
```

Filter by Cluster:

```
| filter kubernetesClusterName == "CLUSTER_NAME"
```

Time-based Analysis:

```
fetch logs, from: now() - DURATION  
| summarize count(), by:{bin(timestamp, INTERVAL)}
```

Top N Pattern:

```
| sort METRIC desc  
| limit N
```

Summary

This notebook provided DQL query patterns for:

- Entity discovery and relationships
 - Container and node metrics
 - Log and event analysis
 - Distributed tracing
 - Multi-source correlation
 - Dashboard visualizations
 - Alerting conditions
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References

- [DQL Documentation](#)
 - [Kubernetes Metrics Reference](#)
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.